Vehicle Maintenance Log — Test Report

Project 1 | October 31, 2026 | Undergraduate Flutter application

Testing summary

20 manual test cases reported as passed (20/20, 100% reported pass rate). Testing was conducted on an Android emulator using the app interface. The results below reflect the tester's submitted outcomes and screenshots; they are not a substitute for automated test logs or an independent audit.

Scope: vehicle CRUD and validation; service-history CRUD and validation; offline persistence and cascading deletes; maintenance calculations and rule-based helper; settings and responsive layout.

Manual test case results

ID / Test	Input or action	Expected result	Actual result / status
T1 — Add valid vehicle	2020 Toyota Camry; 50,000 miles	Vehicle appears with correct fields	PASS (reported) Vehicle listed at 50,000 miles
T2 — Invalid vehicle	Blank make; negative mileage	Form rejects input	PASS (reported) Make and mileage validation messages shown
T3 — Edit vehicle	Change mileage to 52,000	Saved value updates	PASS (reported) 52,000 miles and update confirmation shown
T4 — Cancel vehicle delete	Open confirmation; select Cancel	Vehicle stays in list	PASS (reported) Reported preserved; dialog screenshot available
T5 — Delete vehicle	Delete temporary vehicle	Vehicle removed	PASS (reported) Deletion confirmation message shown
T6 — Add service	Camry oil change at 50,000 miles; $65	Linked service added	PASS (reported) Service shown with $65 and add confirmation
T7 — Invalid service cost	Enter -$10	Input rejected	PASS (reported) Cost validation message shown
T8 — Edit service	Change cost from $65 to $75	Updated value saved	PASS (reported) Service displays $75 and update confirmation
T9 — Delete service	Delete a temporary service record	Record removed	PASS (reported) Delete confirmation shown; original oil change retained
T10 — Offline restart	Close and reopen while offline	Saved data remains	PASS (reported) Reported preserved; dashboard shows 2 vehicles, 1 service, $75
T11 — Cascade deletion	Delete vehicle with linked service	Child service removed, other data kept	PASS (reported) Reported passed; remaining Camry service shown
T12 — Normal maintenance	52,000 current; 50,000 last oil change	3,000 miles remaining	PASS (reported) Planner shows 3,000; next due 55,000
T13 — Boundary warning	54,500 current; 50,000 last service	500 miles left; upcoming warning	PASS (reported) Reported passed
T14 — Overdue maintenance	56,000 current; 50,000 last service	1,000 miles overdue	PASS (reported) Planner displays 1,000 overdue
T15 — No service history	Choose vehicle without past services	Unknown due values; no invented history	PASS (reported) Planner shows no previous service recorded
T16 — Helper counterexample	54,000 current; 1,200 warning threshold	Adjustable Upcoming; fixed-500 On track	PASS (reported) Reported passed
T17 — Dark mode persistence	Enable dark mode; restart	Dark theme remains	PASS (reported) Reported passed
T18 — Hide costs	Turn off Show Service Costs	Dashboard costs hidden	PASS (reported) Reported passed

ID / Test	Input or action	Expected result	Actual result / status
T19 — Dismiss and restore	Dismiss helper; restore suggestion	User can override visibility	PASS (reported) Reported passed
T20 — Responsive view	Rotate emulator to landscape	No overflow; controls usable	PASS (reported) Reported passed

Helper evaluation

Rule-based method: uses recorded service mileage, example service intervals, and a configurable warning distance. Baseline: fixed 500-mile warning. The tested counterexample (T16) uses 54,000 current miles, last oil change at 50,000, and the next estimated service at 55,000. The remaining distance is 1,000 miles; the fixed baseline labels it On track while the adjustable 1,200-mile rule labels it Upcoming. Both are deterministic rules, not a trained model.

Cases T12–T16 cover normal, warning boundary, overdue, no-data, and comparison behavior. Five cases were reported passed. The small hand-chosen sample verifies specified behavior but does not establish real-world predictive accuracy or vehicle-specific service suitability.

Evidence and limitations

App screenshots were supplied for T1–T12, T14 and T15, and additional screenshots for the remaining scenarios were submitted in the testing conversation. Screenshots show visible states; they do not independently prove intermediate actions such as a cancel tap, an offline restart, or a completed cascade deletion. Those actions are recorded as tester-reported results. Keep original screenshots in the repository evidence/ folder with matching test IDs.

No automated test-run results, second-device results, coverage report, or measured peer usability feedback are claimed in this report. Those should be recorded separately after execution. The helper is advisory: example mileage intervals may differ from manufacturer schedules. Users should verify recommendations against their vehicle manual.

Conclusion

All 20 specified manual scenarios were reported as passed. The screenshots support core CRUD, input validation, mileage calculations, and feedback behavior. The app should still receive a final automated test run and an APK installation check before release.